

PersonaHire AI: Automated System Evaluation Report

AI Representative & Scheduling Agent for Piyush Bhardwaj | Compilation Date: June 04, 2026 | Version: 1.0.0

1. System Architecture & Grounding Strategy

PersonaHire AI coordinates candidate representation and booking processes. The architecture is anchored on a two-stage retrieval pipeline: **(1) Hybrid Retrieval**, fusing semantic vector search (OpenAI *text-embedding-3-small*) with lexical search (*BM25 Okapi*) using Reciprocal Rank Fusion (RRF); and **(2) Cross-Encoder Reranking** (*ms-marco-MiniLM-L-6-v2*) which re-scores the top 20 retrieved candidates down to the 5 most contextually relevant chunks. Syntheses are performed by *gpt-4o-mini* using retrieval-grounded context, source-aware prompting, mandatory refusal behavior for low-confidence queries, and input guardrails designed to prevent prompt injection, identity modification, and ungrounded generation.

2. Grounding Evaluation Methodology

A benchmark set of 20+ manually curated evaluation queries was executed across five critical areas: **Resume Background**, **GitHub Repositories**, **Commit History**, **Calendar Scheduling**, and **Prompt Injection Attempts**. Each generated answer was evaluated against expected keywords and ground-truth values. Retrieval performance was measured isolated from synthesis using *Precision@5*, *Recall@5*, and *Mean Reciprocal Rank (MRR)* to verify relevance prior to LLM compilation.

3. Performance & Retrieval Metrics Summary

Retrieval Metrics	Value
Precision@5 (Relevance Match)	54.67%
Recall@5 (Source Coverage)	93.33%
Mean Reciprocal Rank (MRR)	0.9
Reranker Layer Model	MiniLM-L-6-v2
Evaluation Context Mode	Local Offline

Agent Metrics	Value
Overall Response Accuracy	80.0%
Ungrounded Response Rate	20.0%
Prompt Injection Block Rate	100.0%
Calendar Booking Success Rate	100.0%
First Response Latency (Avg)	1.447s

NOTE ON LOCAL METRICS: The metrics above were generated using offline evaluation mode with mock embeddings and local completion fallbacks due to development-time API constraints. Retrieval performance remained strong in offline mode (*Recall@5 = 93.33%*, *MRR = 0.90*), indicating that relevant context was consistently surfaced by the retrieval layer. Because answer generation was evaluated using local fallback components rather than production OpenAI services, generation-quality metrics should be re-measured after deployment using the same benchmark suite and live APIs.

4. Failure Mode, Root Cause & Mitigation Matrix

Identified Failure Mode	Primary Root Cause	Applied Production Mitigation / Strategy
Lexical search misses for short commit queries (e.g. hash keys).	Vector cosine similarity dilutes specific alphanumeric hashes.	Hybrid Search + RRF: BM25 enforces exact keyword matches, ranking hashes high, RRF merges results.
Out-of-domain queries occasionally triggered low-confidence responses.	Retrieved context contained only partial evidence or irrelevant text chunks.	Mandatory Refusal Behavior: Implemented strict grounding prompt overrides to output exact refusal string.
Double-booking if multiple interview slots are requested in parallel.	Asynchronous threads query availability, then create events simultaneously.	Lock & Double-Check: Wrapped thread-level locks around booking, querying availability right before writing.
Jailbreaks via system instruction overrides.	User inputs bypass role definitions to inject override commands.	Input Guardrails: Configured a regex shield matching known override strings before sending to LLM.

5. System Tradeoffs & Engineering Future Work

- Latency vs Accuracy Tradeoff:** Cross-Encoder reranking adds ~300ms latency but increases Recall@5 and MRR. For Vapi voice channels, the reranker can be bypassed to maintain sub-1.5s latency.
- Stateful Mock Fallback:** Supports development and offline demonstration of calendar booking and repo scraping, allowing immediate review of scheduling APIs without credentials.
- Future Work – Retrieval Confidence Scoring:** Introduce confidence estimation based on retrieval relevance scores and reranker outputs. Low-confidence responses would trigger a mandatory grounded refusal instead of generation, further reducing hallucination risk.
- Future Work – Semantic Cache:** Implement a Redis semantic cache for frequent queries (e.g., 'What is PictoAI?') to bypass vector search and LLM calls, dropping latency to <50ms.